Systemfehler – Project Description

Elevator Pitch

Systemfehler (“system failure” or “system error”) is a public-interest open-source initiative aiming to provide barrier-free, multilingual, and easy-to-understand access to information about government services, support programs, and social infrastructure. The project collects scattered, intransparent, and difficult-to-understand data, processes it automatically into structured content, and offers it as a foundation for editorially reviewed and curated information — creating a central hub of knowledge to support people navigating complex life situations.

Background and Motivation

Many social benefits, legal entitlements, and support programs are technically documented somewhere — but their language, structure, and presentation often make them inaccessible. They are difficult to find, especially for people unfamiliar with bureaucratic language or digital tools. Systemfehler addresses this gap by using digital methods to reduce structural exclusion, remove language barriers, and expose intransparency in social systems.

Objectives

The goal of Systemfehler is to automatically collect and analyze publicly available information on social benefits, public services, and institutional access, and present it in a structured, multilingual, and easily understandable format. The automatically extracted data serves only as a starting point — editorial contributions from experts, activists, or native speakers are explicitly desired.

Systemfehler does not compete with existing social organizations or counseling services. It is a complementary digital tool to enhance accessibility and participation.

Technical Approach & Architecture

Systemfehler is based on a modular data processing pipeline, from initial crawling to an accessible frontend. The aim is to handle differently structured content automatically while enabling human review and refinement where needed.

Data Flow and System Components

1. Source Detection

- Identification and categorization of relevant websites (e.g. government agencies, social service providers)

- Configurable crawling jobs regularly fetch new content

2. Data Extraction

- HTML, PDF, or structured sources are parsed

- Key information (title, description, eligibility, target groups) is extracted and segmented

3. Normalization & Structuring

- Unified JSON format based on internal data models

- Language tags, categories, metadata, and context are added

4. Editorial Review

- Marking of potentially inaccessible or complex content (e.g. bureaucratic phrasing)

- Interface for reviewing, simplifying, and translating entries by contributors

5. Output & Interfaces

- Output via API (REST/GraphQL), open data exports, and LLM-compatible formats

- Caching, content validation, and version control for published entries

6. Frontend & Usage

- Multilingual, accessible interface using plain language and filter navigation

- Visualization of categories, rights, exclusions, regional relevance, etc.

- Planned: User feedback features and suggestion system

- Also: Display of current news and topic-specific updates in simple language (e.g. law changes, deadlines, local events)

- Integration or linking of external tools like rent calculators, social assistance estimators, or legal navigation tools

- Display of relevant contact points, forms, and links to application support and counseling services

License and Openness

Systemfehler is fully open source and published under the EUPL v1.2 license. No monetization is planned. The platform is intended to remain low-threshold, accessible, and open to editorial collaboration by third parties.

Development Context

The project is currently in preparation as a part-time side project. Hosting and infrastructure costs are currently covered privately. A sustainable infrastructure would make it possible to invite additional contributors and secure long-term maintenance.

Repository / Code

The codebase is publicly available at: https://github.com/steffolino/systemfehler